# 1 Limitations

The axiom view, and SAI as its formal model, leave out much of what scientific theories are and do. For each limitation we say whether it is partially addressable, and by what, or a genuine limit of a declarative view. The judgements are arguments, not theorems; sentences that rest on a theorem cite it.

**Statistical confrontation with data.** In physics a theory meets data statistically: with models of the instruments and the noise it assigns likelihoods to data, often near-raw (Section 6.3), and statistical theories answer with probabilities. The proof-requiring core of SAI has no native noise model: a datum is proved or not. Graded fit enters only through randomised theories in $M_A$ (Propositions 9.3 and G.3), through the generative variant, whose updating we prove given a support condition and whose identification we only sketch (Proposition A.23 and Remark A.24), or through sentences about probabilities, whose relation to frequencies is not deductive. It matters: on noise-free interval data the syntax of the law texts decides among the forms compatible with the Mars data, while with graded fit the data decide (Table 6 and Example 10.6). So the proof-requiring core is the noise-free special case; its natural extension inside $M_A$ is a law with a calibrated error model; a native treatment is open.

**Written theories and their readers.** A written theory joined to a simple reasoner does not solve the problems the theory is used for; we agree, and see three gaps. *The bridge gap:* a problem posed in ordinary terms becomes formal only through idealisations and bridge principles that the theory does not contain (cf. Cartwright, 1983; Morgan and Morrison, 1999; an extractor prices part of the bridge, Proposition 6.4). *The computational gap:* theory and mathematics suffice logically, but finding the answer needs methods and lemmas (Section 10.5). In the model of Section 7 this gap is a theorem, visible only when what is written, or the time to write it, is charged: program length sees neither reasoner nor vocabulary (Theorem 7.2), written size depends exponentially on both (Corollary 7.10), and charged deduction cannot express the gap (Theorem E.26; derivation sizes sketched in Remark E.27). So insufficiency for a simple reasoner is a property of the pair (theory, reasoner) and of the vocabulary, in the model of Section 7, when what is written, or the time to write it, is charged; this is not a theorem about textbooks, nor about what people carry in their heads (Section 7.9). *The tacit gap:* skills (Polanyi, 1958, 1966) and exemplars (Kuhn, 1970, 1977) are not sentences, and in SAI exemplars are only data: a genuine limit.

**Approximate truth.** "Kepler's laws are only approximately true" is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view as SAI formalises it: its mixture contains randomised theories, and under its pricing an approximately true law is a law with a calibrated error model (Proposition 9.3). The error model is carried by the randomness of the theory machine, not stated as an axiom; a treatment in which it is stated, by sentences about probabilities, belongs to the open problem stated above. Whether a deterministic theory holds a law honestly or with exceptions is a modelling choice of pricing (Proposition G.16); humility is partiality for deterministic theories, calibration for randomised ones (Proposition G.22). A canonical metric for approximate predictions, which we think does not exist, and a general comparative notion of truthlikeness (cf. Popper, 1963; Miller, 1974; Tichý, 1974; Niiniluoto, 1987) are open; our notion is local: a question family, a tolerance, a domain.

| feature of concepts | axiom view | how captured, or what is missing |
|---|---|---|
| for questions and problems, not predictions | yes/no | yes for questions posed in a language (§ 6.3); no for problems needing search or action |
| inert alone; parts of a mind | a limit | worth depends on reasoner and vocabulary (Cor. 7.10); free deduction hides it |
| inferential role constitutive | yes | axioms are inferential relations (meaning postulates, Ramsey–Lewis) |
| compositional | yes/no | yes for contents, as formulas; no for recognizers, which are procedures |
| learning by new links | partly | a persisting symbol, but a global mixture over whole theories |
| soft; meaning change | partly | "within $\varepsilon$ of a sphere" is a true partial theory; not meaning change |
| analogy | partly | partial interpretations, rewarded as they compress (1); not finding them |
| new concepts from old | partly | definitions cost $O(1)$ (Cor. 3.10), help a chaining reasoner (Prop. 7.8) |
| methods attached | partly | lemmas for a bounded reasoner, or axioms via Craig at exponential cost (Lem. F.2); not motor methods |
| recognizers | no | a procedure on situations; the extractor of Prop. 6.4, taken as given |
| one notion, many recognizers | yes | identity is the symbol with its axioms; recognizers are bridges |
| built from pointers | yes | a constant fixed by one axiom, which gathers further axioms |
| purposes | no | axioms can state functions; their use in search is not modelled |
| clarification; facing action | no | not in the inducer |
| natural types | partly | sorts and arities; their emergence is not modelled |

Table 1: Features of human concepts, and whether the axiom view captures them (an assessment).

**Domains, several models, and inconsistent theories in use.** Practice uses mutually inconsistent models of one system, each in its domain (the Earth as flat, as a sphere, as an oblate spheroid), and proof-requiring SAI gives an inconsistent theory weight 0. A sound device exists: honest forms "$q$ lies within $\varepsilon$ of $f$ on the domain" of true models are jointly consistent over a true background, while exact ones can prove everything (Section 9.4). But physicists rarely state or propagate worst-case bounds on the error of an approximate model (measurement uncertainties are propagated routinely, and statistically); worst-case propagation becomes vacuous over long chains; and Lanford's theorem, the rigorous honest form of the Boltzmann equation as described in expositions (Lanford, 1975), holds only for a fraction of the mean free time in the low-density limit, while practice rightly uses the equation far beyond. How to choose a model, a domain or a tolerance is left to tacit understanding; under SAI's pricing a domain is usually a tolerance with an error distribution that varies with the system (Section 9.7), and knowing when a model applies is a recognizer, a genuine limit.

**Concepts.** Table 1 lists features of human concepts that an account of concepts as pieces of code, combined to generate predictions, misses, and whether the axiom view captures them.[1] We accept them as features of human concepts, the first only comparatively (Section 6.1), with two further qualifications: inferential role is one major account of meaning (cf. Sellars, 1953), disputed by referential accounts (cf. Kripke, 1980); and natural types we find plausible. One feature supports the axiom view: a notion with different recognizers has its identity fixed otherwise.

For analogy the universal prior gives a bound: with $K(T)$ the least $K(e)$ such that $\mathsf{W}_e = T$, for every r.e. theory $T_1$ and computable map $\tau$ from sentences to sentences

$$K(T_1 \cup \tau[T_1]) \;\leq\; K(T_1) + K(\tau) + c, \tag{1}$$

since an index of $T_1 \cup \tau[T_1]$ is computable from indices of $T_1$ and $\tau$, and $K(x, y) \leq K(x) + K(y) + c$.

[1] The list follows, with changes, an unpublished draft essay of Hänni's on concepts.

Structure mapping (Gentner, 1983) and models in science (cf. Hesse, 1963) describe such partial interpretations; patchy or stretching concepts (cf. Wilson, 2006; Lakatos, 1976) and interpretation as abduction (Hobbs et al., 1993) are outside the view.

**Other limitations.** *Science predicts very particular things, in set-ups it builds.* SAI captures half, with selected data and partial theories; which questions are worth asking (value-weighted questions would be a handle), and the making of phenomena (cf. Hacking, 1983), are outside it. *Concept formation and discovery.* Step 3 of the ladder (Section 6.1) is partially addressable (Theorem 8.4 and Corollary 7.10); step 4, the revision of the base abstractions, is a genuine limit for now; and SAI evaluates theories in a fixed space of hypotheses without discovering them. *Theory-laden and fallible data.* A base language may presuppose a theory (heliocentric longitudes presuppose Copernican geometry; Definition 10.1; Section 6.1), which biases comparisons between theories that disagree about it; and a deterministic theory must prove even a mislabelled datum, by an exception axiom, for which a noise component in the likelihood, as in our earlier report (Hänni and Claude (Anthropic), 2026), is a partial remedy. *Purposes, technology, and science as a social structure* are genuine limits of a model with one inducer; the last may be where Question 6.1 finds its answer. *The mathematics is free, and large.* Building a background $B'$ into $B$ gains at most $K(B') + c$ bits on any data (Proposition 3.8), a constant that does not grow with the data but is not small for the mathematics physics uses, and that matters when rival theories need different mathematics. More generally, all our verdicts are relative to the universal machine, up to constants that are not small at the scale of real theories (Proposition 3.7). With free deduction, data the background entails carry no information (Proposition 5.15), so induction in mathematics needs a reasoner for whom deduction is costly.

**What the axiom view captures.** It captures the static, declarative and inferential side of theories and concepts (contents, inferential relations and their composition, learning by weighing axioms globally, pointers, partiality, approximate truth in honest or error-model form) and misses the procedural, the statistical and the dynamic side. We offer a model of the declarative core of scientific theories, in which the procedural side reappears formally as the dependence of a written theory's worth on its reasoner and its vocabulary (Section 7).

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